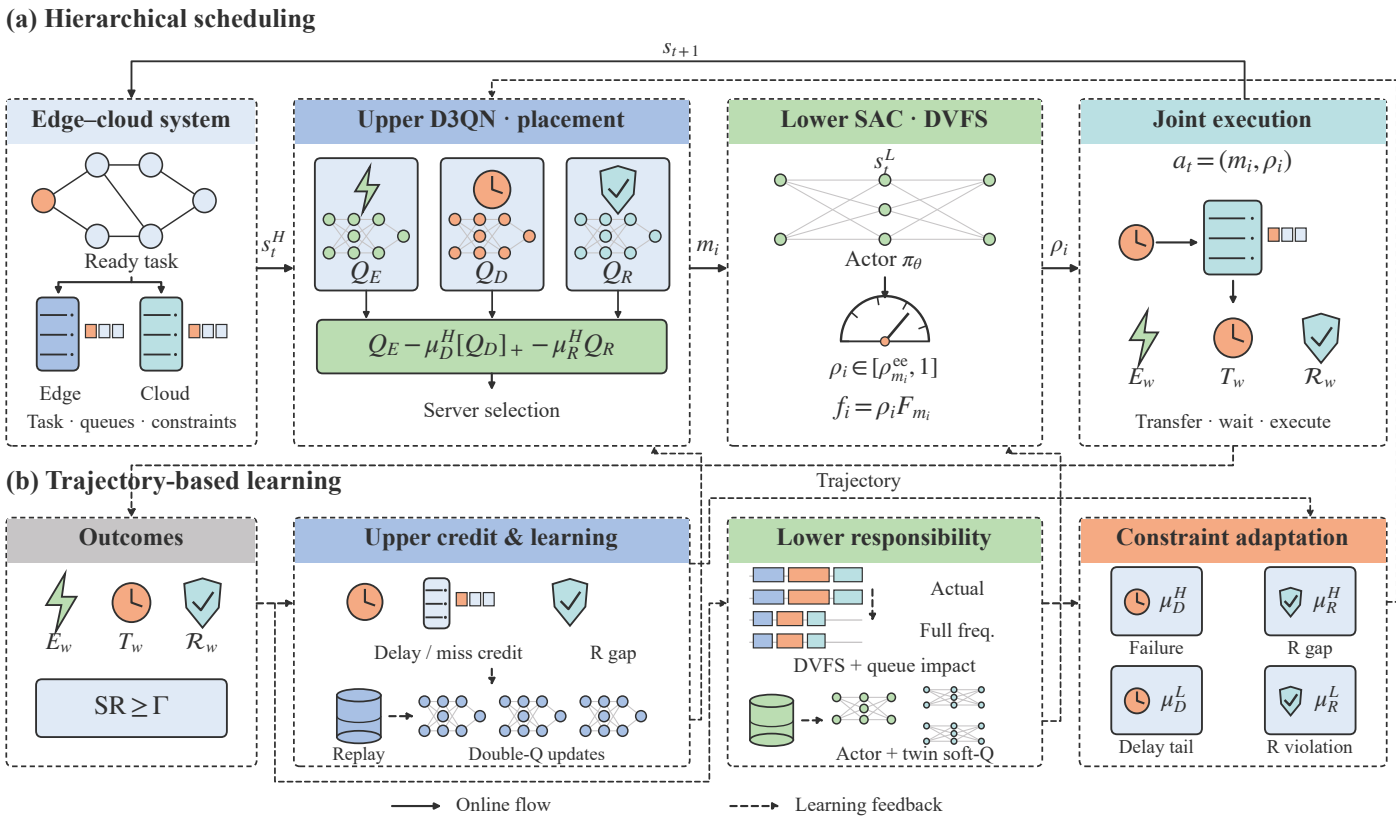

(a) Hierarchical scheduling
s_{t+1}
Edge–cloud system
Ready task
Edge
Cloud
Task · queues · constraints
s_t^H
Upper D3QN · placement
Q_E
Q_D
Q_R
$Q_E - \mu_D^H[Q_D]_+ - \mu_R^H Q_R$
Server selection
m_i
Lower SAC · DVFS
s_t^L
Actor π_θ
$\rho_i \in [\rho_{m_i}^{ee}, 1]$
$f_i = \rho_i F_{m_i}$
ρ_i
Joint execution
$a_t = (m_i, \rho_i)$
E_w
T_w
$\mathcal{R}_w$
Transfer · wait · execute
(b) Trajectory-based learning
Trajectory
Outcomes
E_w
T_w
$\mathcal{R}_w$
SR ≥ Γ
Upper credit & learning
Delay / miss credit
R gap
Replay
Double-Q updates
Lower responsibility
Actual
Full freq.
DVFS + queue impact
Actor + twin soft-Q
Constraint adaptation
μ_D^H
Failure
μ_R^H
R gap
μ_D^L
Delay tail
μ_R^L
R violation
Online flow
Learning feedback